

# Mend-Amar Badral

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Driven by an open mindset with a strong drive for learning new technologies, I aspire to contribute to developing genuine solutions for real world, impactful problems.

## EDUCATION

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**Budapest University of Technology and Economics (BME)** Sep, 2025 – June, 2027  
M.Sc. in Computer Science Engineering  
Budapest, Hungary  
*Specialization:* Data Science & Artificial Intelligence

**National University of Mongolia (NUM)** Sep, 2021 – June, 2025  
B.Sc. in Applied Mathematics  
Ulaanbaatar, Mongolia  
GPA: 3.2/4.0 (rank 3/34)  
*Supervisor:* [Galtbayar Artbazar](#)  
*Thesis:* [Evaluating land degradation using image processing methods](#)

## EXPERIENCE

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**Center of Mathematics Application Research Lab** Oct, 2023 – June, 2025  
*Undergraduate Lab Lead & Research Intern*

- Selected as undergraduate Lab Lead; participated in weekly seminars and collaborative discussions with faculty members on machine learning, statistical modeling, and applied image analysis.
- Developed a custom vendor-specific **Python pipeline** for processing multispectral drone imagery and sensor metadata to generate vegetation and land degradation maps.
- Worked extensively with multispectral image data, vegetation indices (NDVI, MSAVI), raster data processing, and geospatial analysis using Python tools including NumPy, OpenCV, Rasterio, and GDAL.
- Implemented feature engineering and threshold optimization approaches for interpretable land classification and environmental analysis from UAV imagery.

## PROJECTS

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### [Data Integration for 3D Gaussian Splatting](#)

- Implemented a custom JSON parser to extract sensor data obtained by aiSim, an autonomous driving simulator by aiMotive, into a [nerfstudio](#) (3D reconstruction software) format. Performed LiDAR point cloud sensor fusion with the camera images to obtain color information.

### [Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Implemented polynomial regression from Bishop & Bishop's *Deep Learning Foundations and Concepts*, and reproduced key theoretical results in interactive Julia notebooks to empirically analyze **overfitting**, **underfitting**, and the **bias-variance tradeoff**.

## AWARDS

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**Stipendium Hungaricum Scholarship** Sep, 2025

## SKILLS

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**Languages:** English (C1 [IELTS 8.0](#))  
**Programming:** Python, SQL, Bash, JavaScript, Julia, C/C++, R  
**Tools:** Git, Docker, pandas, xarray, sklearn, rasterio, PyTorch, GDAL, OpenCV, nerfstudio  
**Data Engineering:** Large-scale data processing, multi-threading, relational databases (PostgreSQL/MySQL), file formats (GeoTIFF, Parquet, HDF5)